| Reinforcement Learning and Deep Learning | L | P | C |
|---|---|---|---|
| | 3 | | 3 |

| Discipline(s) / EAE / OAE | Semester | Group | Sub-group | Paper Code |
|---|---|---|---|---|
| CSE-AIML | 7 | PC | PC | ML-409T |
| EAE | 7 | AIML-EAE | AIML-EAE-4 | ML-409T |

**Marking Scheme:**

1. Teachers Continuous Evaluation: 25 marks
2. Term end Theory Examinations: 75 marks

**Instructions for paper setter:**

1. There should be 9 questions in the term end examinations question paper.
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.

**Course Objectives :**

| | |
|---|---|
| 1. | To introduce the foundation of Reinforcement learning foundation and Q Network algorithm) |
| 2. | To understand policy optimization ,recent advanced techniques and applications of Reinforcement learning |
| 3. | To introduce the concept of deep learning and neural network |
| 4. | To understand the concept of NLP and computer vision in deep learning |

**Course Outcomes (CO)**

| | |
|---|---|
| **CO 1** | Learn how to define RL tasks and the core principals behind the RL, including policies, value functions, deriving Bellman equations and underst and work with approximate solution(deep Q Network based algorithms) |
| **CO 2** | Learn the policy gradient methods from vanilla to more complex cases and learn application and advanced techniques in Reinforcement Learning |
| **CO 3** | Apply neural networks for problem solving |
| **CO 4** | Able to Analyse images and have basic understanding of NLP in deep learning |

**Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)**

| | PO01 | PO02 | PO03 | PO04 | PO05 | PO06 | PO07 | PO08 | PO09 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO 1** | 3 | 2 | 3 | 3 | 3 | 2 | 2 | - | - | - | - | 2 |
| **CO 2** | 3 | 2 | 3 | 3 | 3 | 2 | 2 | - | - | - | - | 2 |
| **CO 3** | 3 | 2 | 3 | 3 | 3 | 2 | 2 | - | - | - | - | 2 |
| **CO 4** | 3 | 2 | 3 | 3 | 3 | 2 | 2 | - | - | - | - | 2 |

**UNIT-I**

**Reinforcement Learning Foundation:** Introduction to Reinforcement learning and its terms,Features and elements of RL, Defining RL Framework and Markov Decision Process , Polices, Value Functions and Bellman Equations, Exploration vs. Exploitation, Code Standards and Libraries used in RL (Python/Keras/Tensorflow)
**Tabular Methods and Q-networks:** Planning through the use of Dynamic Programming and Monte Carlo, Temporal-Difference learning methods (TD(0), SARSA, Q-Learning), Deep Q-networks (DQN, DDQN, Dueling DQN, Prioritised Experience Replay)

**UNIT-II**

**Policy Optimization:** Introduction to policy-based methods, Vanilla Policy Gradient, REINFORCE algorithm and stochastic policy search, Actor-critic methods (A2C, A3C) ,Advanced policy gradient (PPO, TRPO, DDPG),
**Model-Based RL**: Model-based RL approach
**Recent Advances and Applications:** Meta-learning. Multi-Agent Reinforcement Learning, Partially Observable Markov Decision Process, Applying RL for real-world problems

**UNIT-III**

**Introduction to Deep learning:** Introduction to deep learning and its application,Examples of deep learning
**Introduction to Neural Network:** Introduction to Neural Network its types and application,Introduction to keras,Introduction to ANN Perceptron and its uses,Multilayer perceptron and deep neural network,Activation function and its working TanH function,sigma ,relu etc ,Feed forward network, Cost function, Backpropagation, Gradient Descent, Regulariztion and dropout technique, Batch normalization.
**Types of Neural Network:** Convolutional Neural network,CNN Pooling,CNN Layers,Flattening and Full connection,Preparing a fully connected neural network,Introduction to RNN, Deep RNN, Long Short Term Memory, GRU, Transfer Learning,

**UNIT – IV**

**Deep Learning for Natural Language Processing:** Introduction to NLP and Vector Space Model of Semantics Word Vector Representations: Continuous Skip-Gram Model, Continuous Bag-of-Words model (CBOW), Glove, Evaluations and Applications in word similarity, analogy reasoning
**Deep Learning for Computer Vision:** Image segmentation, object detection, automatic image captioning, Image generation with Generative adversarial networks, video to text with LSTM models. Attention models for computer vision tasks.

**Textbook(s):**
1.Richard S. Sutton and Andrew G. Barto, "Reinforcement learning: An introduction", 2nd Edition, MIT Press, 2019
2. Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. "Deep learning." MIT press, 2016.
3. Antonio Gulli and Sujit Pal,"Deep learning with Keras"

**References:**
1. Wiering, Marco, and Martijn Van Otterlo. "Reinforcement learning: Adaptation, Learning, and Optimization" (2012)
2. Daniel Slater, Gianmario Spacagna and Peter Roelants, "Python Deep Learning", Packt Publication.

| Reinforcement Learning and Deep Learning Lab | L | P | C |
|---|---|---|---|
| | | 2 | 1 |

| Discipline(s) / EAE / OAE | Semester | Group | Sub-group | Paper Code |
|---|---|---|---|---|
| CSE-AIML | 7 | PC | PC | ML-409P |
| EAE | 7 | AIML-EAE | AIML-EAE-4 | ML-409P |

**Marking Scheme:**
1. Teachers Continuous Evaluation: 40 marks
2. Term end Theory Examinations: 60 marks

**Instructions:**
1. The course objectives and course outcomes are identical to that of (Reinforcement Learning and Deep Learning) as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class commencement under intimation to the office of the Head of Department / Institution in which the paper is being offered from the list of practicals below. Atleast 10 experiments must be performed by the students, they may be asked to do more. Atleast 5 experiments must be from the given list.

1. Setting up the Spyder IDE Environment and Executing a Python Program
2. Installing Keras, Tensorflow and Pytorch libraries and making use of them
3. Implement Q-learning with pure Python to play a game
   - Environment set up and intro to OpenAI Gym
   - Write Q-learning algorithm and train agent to play game
   - Watch trained agent play game
4. Implement deep Q-network with PyTorch
5. Python implementation of the iterative policy evaluation and update.
6. Chatbot using bi-directional LSTMs
7. Image classification on MNIST dataset (CNN model with fully connected layer)
8. Train a sentiment analysis model on IMDB dataset, use RNN layers with LSTM/GRU
9. Applying the Deep Learning Models in the field of Natural Language Processing
10. Applying the Convolution Neural Network on computer vision problems

| **Wireless Communication and Networks** | **L** | **P** | **C** |
|---|---|---|---|
| | **3** | | **3** |

| Discipline(s) / EAE / OAE | Semester | Group | Sub-group | Paper Code |
|---|---|---|---|---|
| CSE-NET | 6 | PC | PC | WMC-340T |
| EC-ACT | 6 | PC | PC | WMC-336T |
| EE-VDT | 6 | OAE-ECE-EA | OAE-1 | WMC-336T |
| EAE | 6 | WMC-EAE | WMC-EAE-1C | WMC-336T |
| IT | 7 | PCE | PCE-4 | CIE-415T |

**Marking Scheme:**

1. Teachers Continuous Evaluation: 25 marks
2. Term end Theory Examinations: 75 marks

**Instructions for paper setter:**

1. There should be 9 questions in the term end examinations question paper.
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.

**Course Objectives :**

| | |
|---|---|
| 1. | To know the evolution, principles and architecture of cellular systems and various concepts associated with mobility management and network signaling |
| 2. | To study design and operation of various 2G systems (GSM/GPRS/IS-95/UMTS). |
| 3. | To know features and technical aspects of 3G/4G standards evolved from IMT 2000 vision. |
| 4. | To know technical standards related to 4G, 5G, Bluetooth, Zigbee, NFC, WiMAX. |

**Course Outcomes (CO)**

| | |
|---|---|
| **CO 1** | Understand the evolution, principles and architecture of cellular systems and various concepts associated with mobility management and network signaling. |
| **CO 2** | Develop an understanding of design and operation of various 2G systems (GSM/GPRS/IS-95/CDMA). |
| **CO 3** | Analyze the features and technical aspects of 3G standards evolved from IMT 2000 vision |
| **CO 4** | Analyze the features and technical aspects of 4G, 5G, Bluetooth, Zigbee, NFC and WiMax standards |

**Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)**

| | PO01 | PO02 | PO03 | PO04 | PO05 | PO06 | PO07 | PO08 | PO09 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO 1** | 3 | 1 | 2 | 3 | 3 | 2 | 2 | - | 2 | 2 | 3 | 3 |
| **CO 2** | 3 | 1 | 2 | 2 | 2 | 2 | 3 | - | 3 | 2 | 2 | 3 |
| **CO 3** | 3 | 3 | 3 | 2 | 3 | 2 | 3 | - | 2 | 2 | 3 | 2 |
| **CO 4** | 3 | 3 | 3 | 3 | 3 | 3 | 2 | - | 2 | 3 | 3 | 2 |

**UNIT I**

Introduction To Wireless Communication Systems: Evolution of mobile radio communications; Overview of generations of cellular systems, comparison of various wireless systems.

Introduction to Personal Communication Services (PCS): PCS architecture, Mobility management, Networks signaling. A basic cellular system, multiple access techniques: FDMA, TDMA, CDMA.

Introduction to Wireless Channels and Diversity: Fast Fading Wireless Channel Modeling, Rayleigh/Ricean, Fading Channels, BER Performance in Fading Channels, Introduction to different modulation technique.

**UNIT II**

2G Networks: AMPS, ETACS, GSM: GSM Architecture, Mobility Management, Network signalling, mobile management, voice signal processing and coding, 2.5G Mobile Data Networks: Introduction to Mobile Data Networks, General Packet Radio Services (GPRS): GPRS architecture, GPRS Network nodes, EDGE, Mobile IP,CDMA IS-95: Spread Spectrum, Frequency and Channel Specifications, Forward and Reverse CDMA Channel, Near-Far Problem, Power Control, Spread Spectrum Systems Cellular Code Division Access Systems-Principle, Power Control, effects of multipath propagation on code division multiple access.

**UNIT III**

Third Generation (3G) Mobile Services: The Universsal Mobile Telecommunication System (UMTS) & CDMA 2000 standards, UMTS Network Architecture Release 99, UMTS Interfaces, UMTS Network Evolution, UMTS Channels, UMTS Time Slots.

**UNIT IV**

Evolution of Generations: Features of 4G (LTE & VoLTE), 5G, Bluetooth: Architecture, Feature, Frequency Band, Master-Slave, IEEE Standard, Zigbee: Architecture, Feature, Frequency Band, IEEE Standard, NFC: IEEE Standard, Architecture, Feature, IEEE 802.16 (WiMax): Standard Architecture, Spectrum Allocation, Overview of WiMAX PHY, MAC Layer, Scheduling services, UGS, rtPS, nrtPS, Best Effort (BE).

**Textbook(s):**

1. Raj Pandya, “Mobile & Personnel communication Systems and Services”, Prentice Hall India, 2001.
2. Theodore S. Rappaport, “Wireless Communication- Principles and practices,” 2nd Ed., Pearson Education Pvt. Ltd, 5th Edition, 2008.
3. Wireless Networks: Applications and Protocols, T.S Rappaport, Pearson Education.
4. Wireless Communication and Networks: 3G and Beyond, I. Saha Mishra, TMH Education

**References:**

1. T.L.Singhal “Wireless Communication”, Tata McGraw Hill Publication.
2. Jochen Schiller, “Mobile communications,” Pearson Education Pvt. Ltd., 2002.
3. Yi –Bing Lin & Imrich Chlamatac, “Wireless and Mobile Networks Architecture,” John Wiley & Sons, 2001.
4. Lee, W.C.Y., “Mobile Cellular Telecommunication”, 2nd Edition, McGraw Hill,1998.
5. Smith & Collins, “3G Wireless Networks,” TMH, 2007
6. Schiller, Jochen, “Mobile Communications”, 2nd Edition, Addison Wesley.

| Wireless Communication and Networks Lab | L | P | C |
|---|---|---|---|
| | | 2 | 1 |

| Discipline(s) / EAE / OAE | Semester | Group | Sub-group | Paper Code |
|---|---|---|---|---|
| CSE-NET | 6 | PC | PC | WMC-340P |
| EC-ACT | 6 | PC | PC | WMC-336P |
| EE-VDT | 6 | OAE-ECE-EA | OAE-1 | WMC-336P |
| EAE | 6 | WMC-EAE | WMC-EAE-1C | WMC-336P |
| IT | 7 | PCE | PCE-4 | CIE-415P |

**Marking Scheme:**

1. Teachers Continuous Evaluation: 40 marks
2. Term end Theory Examinations: 60 marks

**Instructions:**

1. The course objectives and course outcomes are identical to that of (Wireless Communication and Networks) as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class commencement under intimation to the office of the Head of Department / Institution in which the paper is being offered from the list of practicals below. Atleast 10 experiments must be performed by the students, they may be asked to do more. Atleast 5 experiments must be from the given list.

1. An introduction to basic digital baseband communication through MATLAB® simulations
2. Simulate Bluetooth voice transmission to observe effect of AWGN and interference of 802.11b on transmission using MATLAB and Simulink.
3. Develop a mobile application for wireless technology using any wizards such as available on www.appypie.com or any other.
4. Simulate the line coding techniques using MATLAB and Simulink
5. Simulate the Binary amplitude shift keying using MATLAB and Simulink
6. Simulate the Binary phase shift keying using MATLAB and Simulink
7. Simulate the Quadrature phase shift keying using MATLAB and Simulink
8. Simulate the Delta Modulation using MATLAB and simulink.
9. Simulate the Adaptive delta Modulation using MATLAB and simulink.
10. Simulate the Direct sequence spread spectrum using MATLAB and Simulink
11. Simulate WSN node to determine position on node and blink LED using cupcarbon simulator and senscript.

| Web Mining | L | P | C |
|---|---|---|---|
| | 3 | | 3 |

| Discipline(s) / EAE / OAE | Semester | Group | Sub-group | Paper Code |
|---|---|---|---|---|
| CSE/IT | 7 | PCE | PCE-5 | CIE-431T |

**Marking Scheme:**

1. Teachers Continuous Evaluation: 25 marks
2. Term end Theory Examinations: 75 marks

**Instructions for paper setter:**

1. There should be 9 questions in the term end examinations question paper.
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.

**Course Objectives :**

| | |
|---|---|
| 1. | To understand the scope of Web mining, identifying the opportunities and the challenges |
| 2. | To learn to apply different data mining/ML techniques in web mining |
| 3. | To learn graph based representation of WWW |
| 4. | To understand techniques for web crawling for web contents to build useful statistics like page ranking |

**Course Outcomes (CO)**

| | |
|---|---|
| **CO 1** | Understand the scope of Web mining, identifying the opportunities and the challenges |
| **CO 2** | Learn to apply different data mining/ML techniques in web mining |
| **CO 3** | Learn graph based representation of WWW |
| **CO 4** | Understand techniques for web crawling for web contents to build useful statistics like page ranking |

**Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)**

| | PO01 | PO02 | PO03 | PO04 | PO05 | PO06 | PO07 | PO08 | PO09 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO 1** | 3 | 2 | - | - | - | - | - | - | - | - | - | 2 |
| **CO 2** | 2 | 2 | 3 | 1 | - | - | - | - | - | - | - | 2 |
| **CO 3** | 2 | 2 | 2 | - | - | - | - | - | - | - | - | 2 |
| **CO 4** | 3 | 2 | 2 | 1 | 3 | - | - | - | - | - | - | 2 |

**UNIT-I**

World Wide Web – Data Mining Vs Web Mining – Data Mining Foundations: Association rules and Sequential Patterns – Machine Learning in Data Mining, Web Mining: Web Structure Mining, Web Content Mining, and Web Usage Mining. Web Structure Mining: Web Graph - Extracting pattern from hyperlinks – Mining Document Structure - PageRank.

**UNIT-II**

Web Content Mining: Text and Web Page Pre-processing – Inverted Indices – Latent Semantic Indexing – Web Spamming – Social Network Analysis – Web Crawlers – Structured Data Extraction – Opinion mining and Sentiment Analysis.

**UNIT-III**

Web usage Mining: Data collection and Pre-processing – Data Modelling – Discovery and Analysis of Web Usage – Recommender System and Collaborative Filtering – Query log mining

**UNIT – IV**

**Web Mining Applications and Other Topics:** Data integration for e-commerce, Web personalization and recommender systems, Web content and structure mining, Web data warehousing, Review of tools, applications, and systems

**Textbook(s):**

1. Liu B. Web Data Mining: Exploring Hyperlinks, Contents, and Usage Data. Springer Science & Business Media; 2007.

**References:**

1. Markov Z, Larose DT. Data Mining the Web: Uncovering Patterns in Web Content, Structure, and Usage. John Wiley & Sons; 2007.
2. Web Mining:: Applications and Techniques by Anthony Scime
3. Mining the Web: Discovering Knowledge from Hypertext Data by Soumen Chakrabarti

| Web Mining Lab | L | P | C |
|---|---|---|---|
| | | 2 | 1 |

| Discipline(s) / EAE / OAE | Semester | Group | Sub-group | Paper Code |
|---|---|---|---|---|
| CSE/IT | 7 | PCE | PCE-5 | CIE-431P |

**Marking Scheme:**
1. Teachers Continuous Evaluation: 40 marks
2. Term end Theory Examinations: 60 marks

**Instructions:**
1. The course objectives and course outcomes are identical to that of (Web Mining) as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class commencement under intimation to the office of the Head of Department / Institution in which the paper is being offered from the list of practicals below. Atleast 10 experiments must be performed by the students, they may be asked to do more. Atleast 5 experiments must be from the given list.

1. Implement Page Rank Algorithm in Web Mining
2. Analyze the link structure of web using page rank algorithms.
3. Text and webpage pre-processing
4. Social network analysis
5. Opinion mining
6. Sentiment analysis
7. Privatization of web content
8. Web usage mining
9. Recommender system
10. Web structure mining

| Machine Learning | L | P | C |
|---|---|---|---|
| | 3 | | 3 |

| Discipline(s) / EAE / OAE | Semester | Group | Sub-group | Paper Code |
|---|---|---|---|---|
| ECE | 6 | PCE | PCE-3 | ECE-350T |
| EAE | 6 | MLDA-EAE | MLDA-EAE-2C | ML-342T |
| CSE/IT/CST/ITE | 7 | PCE | PCE-5 | CIE-421T |
| CSE-AIML | 7 | PC | PC | ML-407T |
| EAE | 7 | AIML-EAE | AIML-EAE-3 | ML-407T |

**Marking Scheme:**

1. Teachers Continuous Evaluation: 25 marks
2. Term end Theory Examinations: 75 marks

**Instructions for paper setter:**

1. There should be 9 questions in the term end examinations question paper.
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.

**Course Objectives :**

| | |
|---|---|
| 1. | To understand the need of machine learning |
| 2. | To learn about regression and feature selection |
| 3. | To understand about classification algorithms |
| 4. | To learn clustering algorithms |

**Course Outcomes (CO)**

| | |
|---|---|
| **CO 1** | To formulate machine learning problems |
| **CO 2** | Learn about regression and feature selection techniques |
| **CO 3** | Apply machine learning techniques such as classification to practical applications |
| **CO 4** | Apply clustering algorithms |

**Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)**

| | PO01 | PO02 | PO03 | PO04 | PO05 | PO06 | PO07 | PO08 | PO09 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO 1** | 3 | 3 | 3 | 3 | 3 | 2 | 2 | - | - | - | - | 2 |
| **CO 2** | 3 | 3 | 3 | 3 | 3 | 2 | 2 | - | - | - | - | 2 |
| **CO 3** | 3 | 3 | 3 | 3 | 3 | 2 | 2 | - | - | - | - | 2 |
| **CO 4** | 3 | 3 | 3 | 3 | 3 | 2 | 2 | - | - | - | - | 2 |

**UNIT-I**

**Introduction**: Machine learning, terminologies in machine learning, Perspectives and issues in machine learning, application of Machine learning, Types of machine learning: supervised, unsupervised, semi-supervised learning. Review of probability, Basic Linear Algebra in Machine Learning Techniques, Dataset and its types,Data preprocessing, Bias and Variance in Machine learning , Function approximation, Overfitting

**UNIT-II**

**Regression Analysis in Machine Learning**: Introduction to regression and its terminologies,Types of regression,Logistic Regression

**Simple Linear regression**: Introduction to Simple Linear Regression and its assumption, Simple Linear Regression Model Building,Ordinary Least square estimation, Properties of the least-squares estimators and the fitted regression model, Interval estimation in simple linear regression , Residuals
**Multiple Linear Regression**:Multiple linear regression model and its assumption, **Interpret Multiple Linear Regression Output(R-Square, Standard error, F, Significance F, Cofficient P values), Access the fit of multiple linear regression model (**R squared, Standard error)
**Feature Selection and Dimensionality Reduction**: PCA, LDA, ICA

**UNIT-III**

**Introduction to Classification and Classification Algorithms:** What is Classification? General Approach to Classification, k-Nearest Neighbor Algorithm, Random Forests, Fuzzy Set Approaches
**Support Vector Machine**: Introduction, Types of support vector kernel – (Linear kernel, polynomial kernel, and Gaussiankernel), Hyperplane – (Decision surface), Properties of SVM, and Issues in SVM.
**Decision Trees**: Decision tree learning algorithm,ID-3algorithm, Inductive bias, Entropy and information theory, Information gain,Issues in Decision tree learning.
**Bayesian Learning** - Bayes theorem, Concept learning, Bayes Optimal Classifier, Naïve Bayes classifier, Bayesian belief networks, EM algorithm
**Ensemble Methods**: Bagging, Boosting and AdaBoost and XBoost,
**Classification Model Evaluation and Selection:** Sensitivity, Specificity, Positive Predictive Value, Negative Predictive Value, Lift Curves and Gain Curves, ROC Curves, Misclassification Cost Adjustment to Reflect Real-World Concerns, Decision Cost/Benefit Analysis

**UNIT – IV**

**Introduction to Cluster Analysis and Clustering Methods:** The Clustering Task and the Requirements for Cluster Analysis , Overview of Some Basic Clustering Methods:-k-Means Clustering, k-Medoids Clustering, Density-Based Clustering: DBSCAN - Density-Based Clustering Based on Connected Regions with High Density, Gaussian Mixture Model algorithm , Balance Iterative Reducing and Clustering using Hierarchies (BIRCH) , Affinity Propagation clustering algorithm,Mean-Shift clustering algorithm, ordering Points to Identify the Clustering Structure (OPTICS) algorithm, Agglomerative Hierarchy clustering algorithm, **Divisive Hierarchical** , Measuring Clustering Goodness

**Textbook(s):**

1. Tom M. Mitchell, “Machine Learning”, McGraw-Hill Education (India) Private Limited, 2013.
2. M. Gopal, “Applied Machine Learning”, McGraw Hill Education

**References:**
1. C. M. BISHOP (2006), “Pattern Recognition and Machine Learning”, Springer-Verlag New York, 1st Edition
2. R. O. Duda, P. E. Hart, D. G. Stork (2000), Pattern Classification, Wiley-Blackwell, 2nd Edition

| Machine Learning Lab | L | P | C |
|---|---|---|---|
| | | 2 | 1 |

| Discipline(s) / EAE / OAE | Semester | Group | Sub-group | Paper Code |
|---|---|---|---|---|
| ECE | 6 | PCE | PCE-3 | ECE-350P |
| EAE | 6 | MLDA-EAE | MLDA-EAE-2C | ML-342P |
| CSE/IT/CST/ITE | 7 | PCE | PCE-5 | CIE-421P |
| CSE-AIML | 7 | PC | PC | ML-407P |
| EAE | 7 | AIML-EAE | AIML-EAE-3 | ML-407P |

**Marking Scheme:**
1. Teachers Continuous Evaluation: 40 marks
2. Term end Theory Examinations: 60 marks

**Instructions:**
1. The course objectives and course outcomes are identical to that of (Machine Learning) as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class commencement under intimation to the office of the Head of Department / Institution in which the paper is being offered from the list of practicals below. Atleast 10 experiments must be performed by the students, they may be asked to do more. Atleast 5 experiments must be from the given list.

1. Introduction to JUPYTER IDE and its libraries Pandas and NumPy
2. Program to demonstrate Simple Linear Regression
3. Program to demonstrate Logistic Regression
4. Program to demonstrate Decision Tree – ID3 Algorithm
5. Program to demonstrate k-Nearest Neighbor flowers classification
6. Program to demonstrate Naïve- Bayes Classifier
7. Program to demonstrate PCA and LDA on Iris dataset
8. Program to demonstrate DBSCAN clustering algorithm
9. Program to demonstrate K-Medoid clustering algorithm
10. Program to demonstrate K-Means Clustering Algorithm on Handwritten Dataset

| Pattern Recognition and Computer Vision | L | P | C |
|---|---|---|---|
| | 3 | | 3 |

| Discipline(s) / EAE / OAE | Semester | Group | Sub-group | Paper Code |
|---|---|---|---|---|
| CSE-AIML | 7 | PC | PC | ML-411T |
| EAE | 7 | AIML-EAE | AIML-EAE-5 | ML-411T |

**Marking Scheme:**

1. Teachers Continuous Evaluation: 25 marks
2. Term end Theory Examinations: 75 marks

**Instructions for paper setter:**

1. There should be 9 questions in the term end examinations question paper.
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.

**Course Objectives :**

| | |
|---|---|
| 1. | Understand the in-depth concept of Pattern Recognition |
| 2. | Implement Bayes Decision Theory |
| 3. | Understand the in-depth concept of Perception and related Concepts |
| 4. | Understand the concept of ML Pattern Classification |

**Course Outcomes (CO)**

| | |
|---|---|
| **CO 1** | Discuss various concepts of pattern recognition |
| **CO 2** | Understanding various algorithms |
| **CO 3** | Explain and apply various computer vision techniques |
| **CO 4** | Describe the concept of shape analysis and filtering |

**Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)**

| | PO01 | PO02 | PO03 | PO04 | PO05 | PO06 | PO07 | PO08 | PO09 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO 1** | 3 | 2 | 2 | 3 | 3 | - | 2 | - | - | 1 | 3 | 2 |
| **CO 2** | 3 | 3 | 1 | 1 | 1 | - | 1 | 1 | - | 2 | 2 | 1 |
| **CO 3** | 3 | 2 | 3 | 3 | 2 | - | 2 | - | - | 2 | 3 | 1 |
| **CO 4** | 1 | 2 | 3 | 2 | 2 | - | 1 | - | - | 1 | 2 | 2 |

**UNIT-I**

Induction Algorithms. Rule Induction. Decision Trees. Bayesian Methods. The Basic Naıve Bayes Classifier. Naive Bayes Induction for Numeric Attributes. Correction to the Probability Estimation. Laplace Correction. No Match. Other Bayesian Methods. Other Induction Methods. Neural Networks. Genetic Algorithms. Instance-based Learning. Support Vector Machines.

**UNIT-II**

About Statistical Pattern Recognition. Classification and regression. Features and Feature Vectors, and Classifiers. Pre-processing and feature extraction. The curse of dimensionality. Polynomial curve fitting. Model complexity. Multivariate non-linear functions. Bayes' theorem. Decision boundaries. Parametric methods. Sequential parameter estimation. Linear discriminant functions. Fisher's linear discriminant. Feed-forward network mappings.

**UNIT-III**

Review of image processing techniques – classical filtering operations – thresholding techniques – edge detection techniques – corner and interest point detection – mathematical morphology – texture.

**UNIT – IV**

Binary shape analysis – connectedness – object labelling and counting – size filtering – distance functions – skeletons and thinning – deformable shape analysis – boundary tracking procedures – active contours – shape models and shape recognition – centroidal profiles – handling occlusion – boundary length measures – boundary descriptors – chain codes – Fourier descriptors – region descriptors – moments.

**Textbook(s):**

1. Pattern Classification, Richard O. Duda, Peter E. Hart, and David G. Stork. Wiley, 2000, 2nd Edition
2. D. L. Baggio et al., Mastering OpenCV with Practical Computer Vision Projects, Packt Publishing, 2012.

**References:**

1. Pattern Recognition, Jürgen Beyerer, Matthias Richter, and Matthias Nagel. 2018
2. E. R. Davies, Computer & Machine Vision, Fourth Edition, Academic Press, 2012

| Pattern Recognition and Computer Vision Lab | L | P | C |
|---|---|---|---|
| | | 2 | 1 |

| Discipline(s) / EAE / OAE | Semester | Group | Sub-group | Paper Code |
|---|---|---|---|---|
| CSE-AIML | 7 | PC | PC | ML-411P |
| EAE | 7 | AIML-EAE | AIML-EAE-5 | ML-411P |

**Marking Scheme:**
1. Teachers Continuous Evaluation: 40 marks
2. Term end Theory Examinations: 60 marks

**Instructions:**
1. The course objectives and course outcomes are identical to that of (Pattern Recognition and Computer Vision) as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class commencement under intimation to the office of the Head of Department / Institution in which the paper is being offered from the list of practicals below. Atleast 10 experiments must be performed by the students, they may be asked to do more. Atleast 5 experiments must be from the given list.

1. Write a MATLAB/Python function that computes the value of the Gaussian distribution N(m,s) at given vector X and plot the effect of varying mean and variance to the normal distribution.
2. Implementation of Gradient descent.
3. Implementation of Linear Regression using Gradient descent.
4. Comparison of classification accuracy of SVM and CNN for the dataset.
5. Implementation basic Image Handling and processing operations on the image.
6. Implementation of Geometric Transformation.
7. Implementation of Perspective Transformation.
8. Implementation of Camera Calibration
9. Compute Fundamental Matrix.

| Principles of Entrepreneurship Mindset | L | P | C |
|---|---|---|---|
| | 2 | | 2 |

| Discipline(s) / EAE / OAE | Semester | Group | Sub-group | Paper Code |
|---|---|---|---|---|
| All | 7 | HS/MS | MS | MS-401 |

**Marking Scheme:**

1. Teachers Continuous Evaluation: 25 marks
2. Term end Theory Examinations: 75 marks

**Instructions for paper setter:**

1. There should be 9 questions in the term end examinations question paper.
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.

**Course Objectives :**

| | |
|---|---|
| 1. | To understand basic aspects of establishing a business in a competitive environment |
| 2. | To apply the basic understanding to examine the existing business ventures |
| 3. | To examine various business considerations such as marketing, financial and teaming etc. |
| 4. | To assess strategies for planning a business venture |

**Course Outcomes (CO)**

| | |
|---|---|
| **CO 1** | Understand basic aspects of establishing a business in a competitive environment |
| **CO 2** | Apply the basic understanding to examine the existing business ventures |
| **CO 3** | Examine various business considerations such as marketing, financial and teaming etc. |
| **CO 4** | Assessing strategies for planning a business venture |

**Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)**

| | PO01 | PO02 | PO03 | PO04 | PO05 | PO06 | PO07 | PO08 | PO09 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO 1** | 2 | 2 | 1 | 2 | - | 2 | - | - | 1 | 2 | 3 | 2 |
| **CO 2** | 2 | 2 | 1 | 2 | - | 2 | - | - | 1 | 2 | 3 | 2 |
| **CO 3** | 2 | 2 | 1 | 2 | - | 2 | - | - | 1 | 2 | 3 | 2 |
| **CO 4** | 2 | 2 | 1 | 2 | - | 2 | - | - | 1 | 2 | 3 | 2 |

**UNIT-I**

**Entrepreneurial perspective:** Foundation, Nature and development ofentrepreneurship, importance of entrepreneurs,Entrepreneurial Mind, Individual entrepreneur Typesof entrepreneurs, Entrepreneurship in India

**UNIT-II**

**Beginning Considerations:** Creativity and developing business ideas; Creatingand starting the venture; Building a competitiveadvantage; Opportunity recognition, Opportunityassessment; Legal issues

**UNIT-III**

**Developing Financial Plans:** Sources of Funds,Managing Cash Flow,Creating a successful Financial PlanDeveloping a business plan

**UNIT - IV**

**Developing Marketing Plans:** Developing a powerful Marketing Plan, E-commerce,Integrated Marketing Communications
**Leading Considerations:** Developing Team, Inviting candidates to join team,Leadership model

**Textbook(s):**

1. Robert D Hisrich, Michael P Peters & Dean A Shepherd, "Entrepreneurship" 10th Edition,McGraw Hill Education, 2018

**References:**

1. Norman M. Scarborough and Jeffery R. cornwell, "Essentials of entrepreneurship and small business management" 8th Edition, Pearson, 2016
2. Rajiv Roy, "Entrepreneurship", 2nd Edition, Oxford University Press, 2011
3. Sangeeta Sharma, "Entrepreneurship Development", 1st Edition, Prentice-Hall India, 2016
4. John Mullins, "The New Business Road Test: What entrepreneurs and investors should dobefore launching a lean start-up" 5th Edition, Pearson Education, 2017
5. Charantimath, Entrepreneurship Development and Small Business Enterprise, Pearson Education.